

Arcesium Intern Interview Experience 2022

About the company

Global financial technology and professional services firm, delivering solutions to some of the world's most sophisticated financial institutions. Launched by D.E. Shaw group with additional equity backing from Blackstone Alternative Asset Management and J.P. Morgan.

Online Assessment (Platform : HackerRank)

- Quantitative Aptitude - 15 MCQ's (20 minutes)
- Technical Aptitude - 15 MCQ's (15 minutes)
- Coding - 2 problems (45 minutes)

For Quantitative Aptitude, refer indiabix.com. Problems were asked which made use of commonly asked formulas (profit/loss, family members relations and ages, train crossing, etc.). The only drawback is that the questions would have long workouts and insufficient time. So don't go for high precision accuracy and waste time. Use tricks such as rounding off percentages as the choices are given such that they can be eliminated (i.e., the choices have significant differences between them). Also, there's negative marking of -0.25. But if you solve like about 8-10 questions then you can go with random choices for the remaining questions as in the worst case you might get a -1.75 to the total score.

For Technical Aptitude, the questions were a mixed bag - output questions from C, C++ and Java, and a few DBMS queries as well. Same marking scheme as prior section: +1/-0.25. Try to be sure about the syntax in at least two of the three languages and make sure to be thorough with precedence of operators and oops concepts such as polymorphism and virtual classes, since these are applied in the questions that were asked.

For coding, the following problems were asked -

1. Two players are playing a game which is as follows: There is a heap of n stones. In one move a player can remove a number of stones which is a power of 2. Whoever removes the last stone wins the game. Given n , print the winning player number (1 or 2).

Solution: Basically, we have to find the number of set bits in the given n . If it's odd, print 1. Else print 2. But the catch was the range of integers was 0 to 10^{250} . There's a class called BigInteger in Java under `java.math` to handle very large integers.

Here's the code:

```
import java.util.*;
import java.math.*;

class Solution
{
    public static void main (String[] args)
    {
        Scanner s = new Scanner(System.in);
        String str = new String();
```

```

        str = s.nextLine();
        BigInteger b = new BigInteger(str);
        if (b.bitCount() % 2 == 1)
            System.out.println("1");
        else
            System.out.println("2");
    }
}

```

The probability of getting such questions where you are asked to manipulate very large integers is very low. But still being familiar with BigInteger would help you a lot.

2. The question was similar to the one given below:

<https://www.geeksforgeeks.org/maximum-number-of-teams-of-size-k-possible-with-each-player-from-different-country/>

As Arcesium is a Fin-Tech company, they give more weightage to the aptitude section. So, make sure to solve at least 50% of the questions in the first two sections and then rock the coding questions :).

Fifteen students were selected for the subsequent Interview rounds.

Round 1 - Code Pair

I was asked to implement a binary tree and then find the ancestors of a given target node. After completing that, I was asked to optimize the code.

[Code Pair round has a factor of luck since each one would get a different interviewer. From what I heard, many of us got relatively easy questions from binary trees and linked lists. But one guy was asked to do division without using multiplication, division or mod operation, which had to be done using bit manipulation.

(Link: <https://www.geeksforgeeks.org/divide-two-integers-without-using-multiplication-division-mod-operator/>)

So, try to solve lots of standard easy and medium problems from LeetCode and be thorough with the basic question patterns and expect the very best to happen.

(Reference Links:

<https://leetcode.com/discuss/general-discussion/460599/blind-75-leetcode-questions>

<https://hackernoon.com/14-patterns-to-ace-any-coding-interview-question-c5bb3357f6ed>)]

Round 2 - Technical Interview I

The interviewer and I exchanged intros and then he asked me what my favourite subject was. I told him OS since I was strong in it. (You can answer whatever subject you feel strong enough in but mostly they would try to cover questions from all subjects - OS, DBMS, OOPS). He asked me about bootloader program and then about four pillars of OOPS. Then he proceeded with the difference

between Run-time and Compile-time polymorphism. Run-time is slower as compared to compile-time. He asked me the reason and the answer was VTABLE lookup. So, make sure to be thorough with concept of abstract class and virtual functions. Then he asked me about caching.

Then a few common questions such as what is an API and what happens when you type google.com in your browser. Then he asked the difference between HTTP and HTTPS and about SSL. (These were general questions and were particular to this interviewer. In general, you don't need to worry much if you get these types of questions. Just manage to answer them to the best of your capability).

Then he asked me to explain about the relation schema of my DBMS project. He asked me, "Do you see the problem here?" (I had two tables which contained similar types of data) So I answered that we can merge them into one. He asked me about normalization and then proceeded with "Do you see the problem here" again. Apparently on merging the two tables the API calls get made to the same table twice, which would lead to slower performance. He asked me for the solution and the answer was to use Indices. Then he asked questions which data structure is used to implement indexing and the difference between caching and indexing and then when to use indexing. So be thorough with the relation schemas of your project and learn about SQL indices, normalization and queries. They might not be asked directly but as part of questioning about your project.

Round 3 - Technical Interview II

The Director of Engineering had a talk with me and gave me a coding problem:

Given a stream of posts like Instagram Feed, you have to pair posts which were 25 seconds apart in their timestamps. My idea was to use a queue and then remove the head of the queue as long as the difference was greater than 25. Then if the head was exactly 25 seconds less than the current post, make pair. I was asked only to write pseudocode. After that, another interviewer started asking me a stream of OS questions. He covered the following topics:

- Program vs Process
- Scheduling algorithms
- Thread vs Process
- States of a process
- Semaphore vs Mutex
- Deadlocks

After these three rounds, I attended an HR round, where I was asked basic HR questions.

Result: Three of us got selected.

Do check out past intern interview experiences from GFG and practise company specific question sets too. All the best :)

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